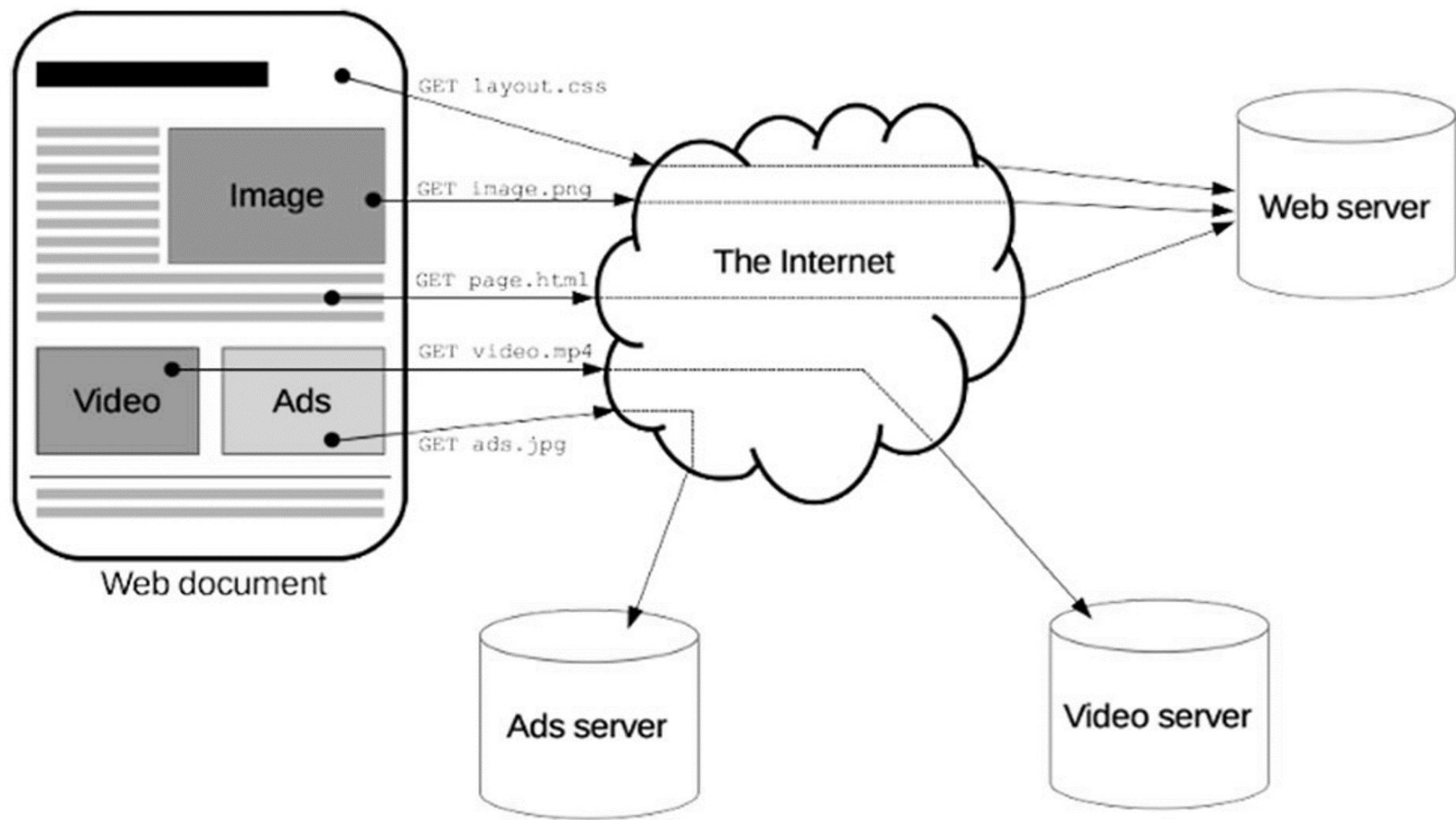


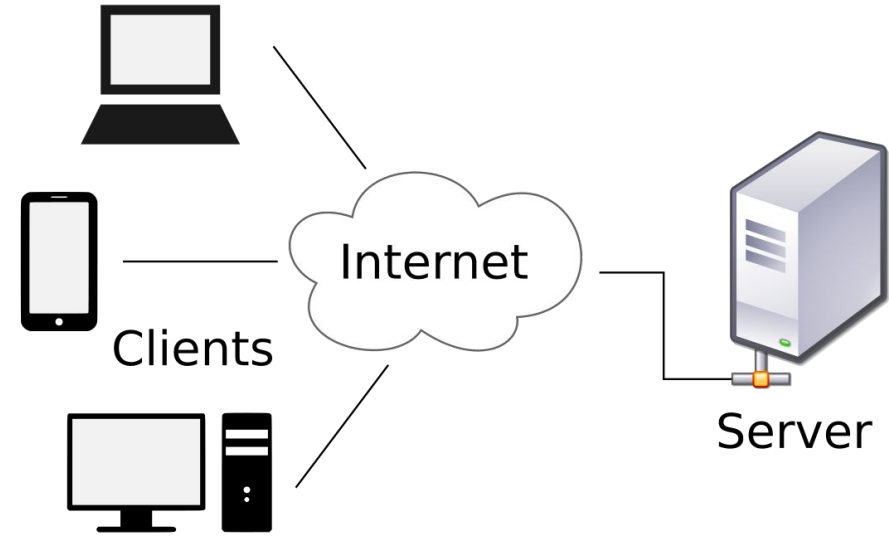
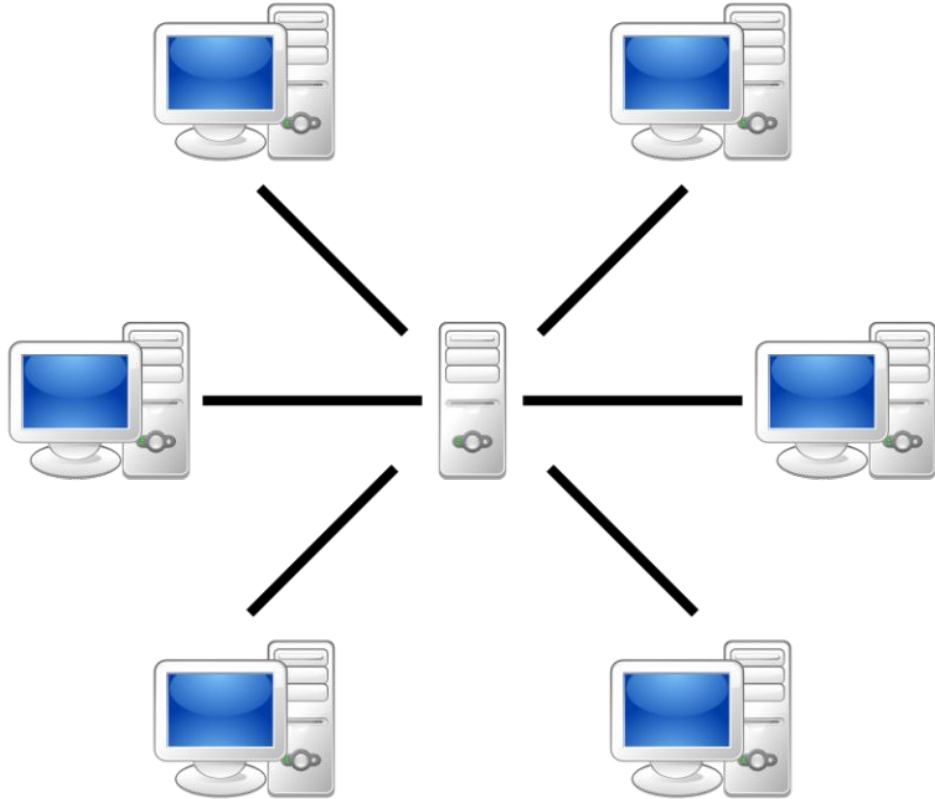
Web Fundamentals

Introduction to WEB

- The web has, from its inception, been structured around the idea of resources.
- Generally a platform for sharing simple text/HTML based files, documents, images etc. (Resource-oriented)
- Most web applications are built around a client-server model.
- Client could be something as simple as a web browser, programs, scripts.
- Server presents only as a single machine virtually.

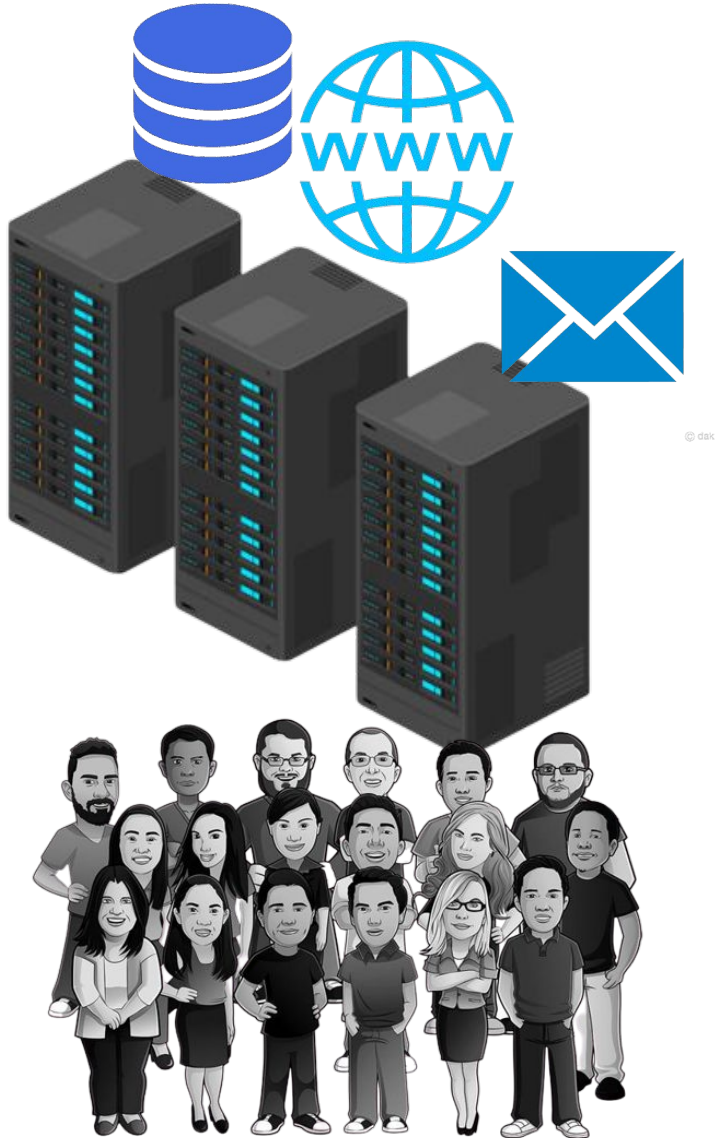


Types of server

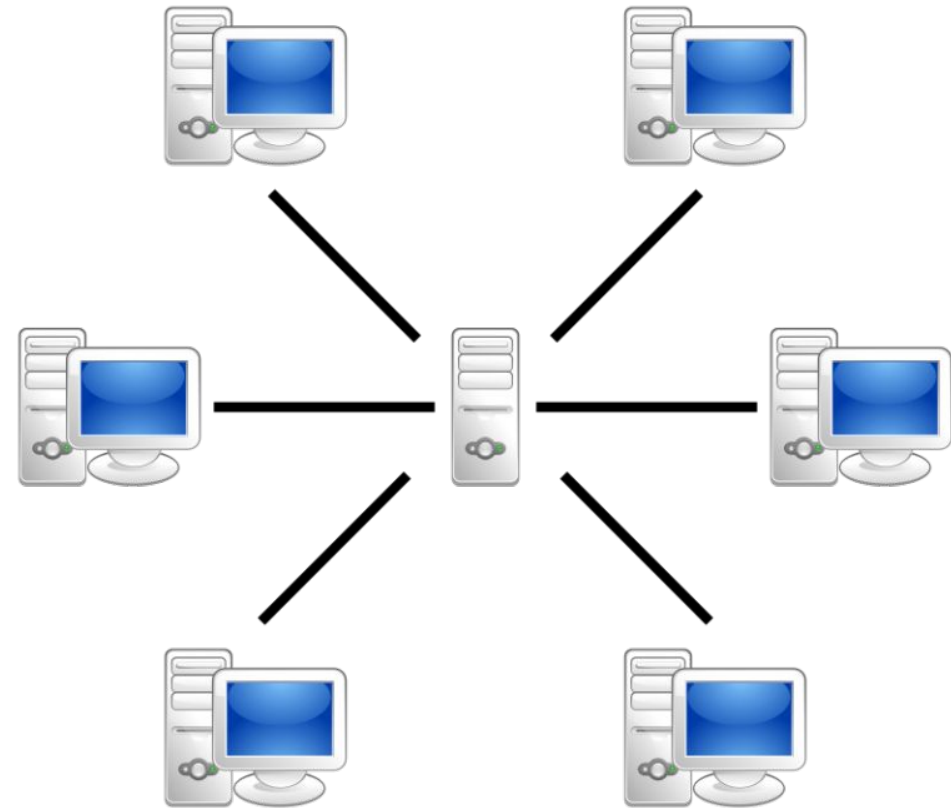


Types of server

Dedicated server for each service
(for large organization)

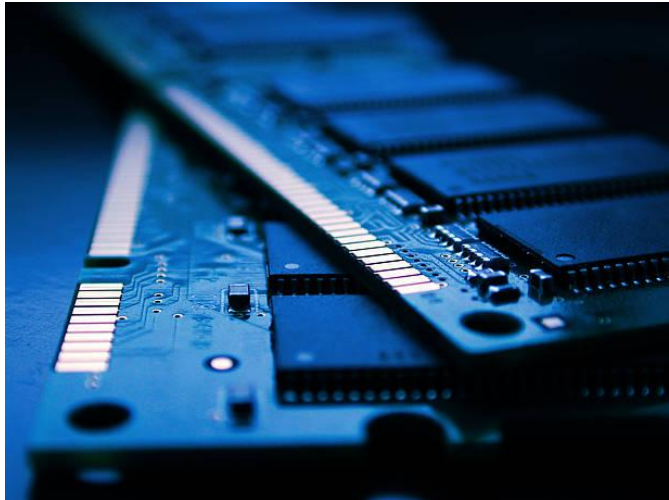
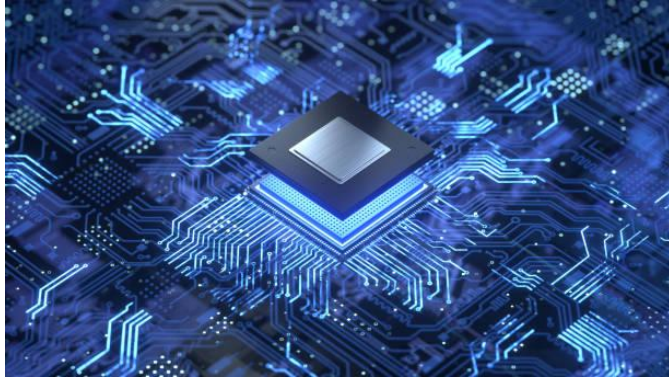


Single dedicated server handle all task
(for small organization)

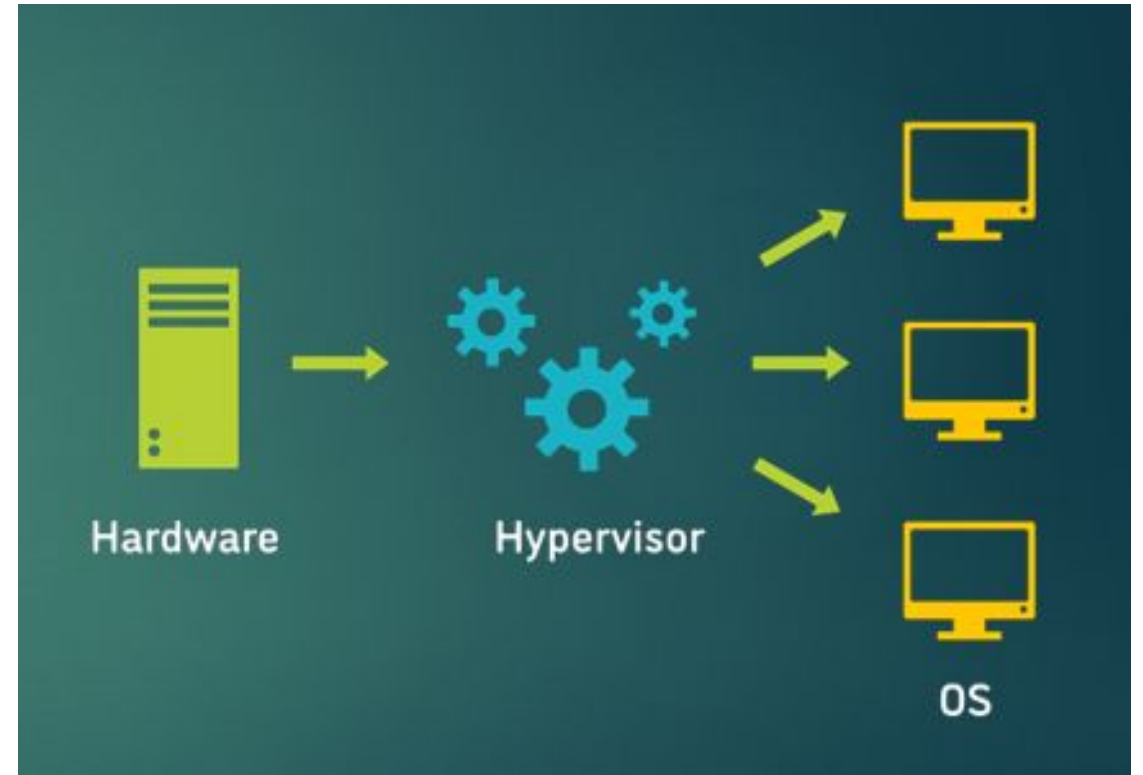


Types of server

Physical



Virtual

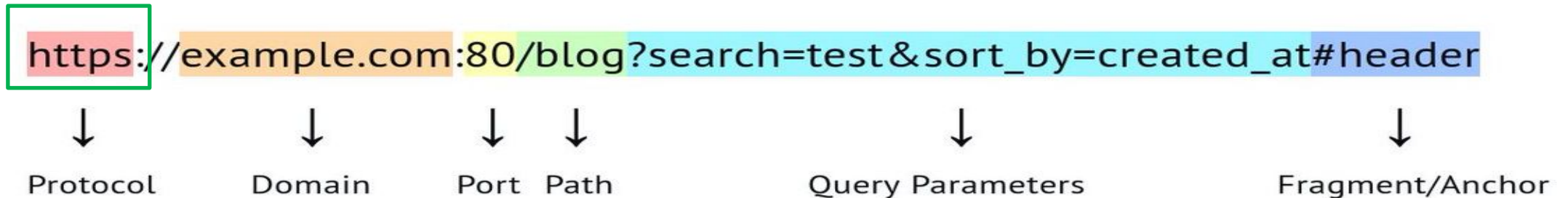


**But How these messages are
transferred?**

URL

1. Stands for Uniform Resource Locator
2. URL refers to the physical location of the resource to be fetched.
"/home.htm!"
3. URI (uniform Resource Identifier): refers to any resource that needs to be fetched.
 - It may be a physical resource or not. ("/users", "/users/user1/posts")

URL Anatomy



Protocol?

What is protocol and why we need it?

1. Internet Protocols are a set of rules that governs the communication and exchange of data over the internet. Both the sender and receiver should follow the same protocols in order to communicate the data
2. This protocol allows a web server to:
 1. Receive the information-requests-sent by any client.
 2. Process these requests..
 3. Send the appropriate responses.
3. There are lots of protocols that can transfer data between client and a server.
 1. Ex: HTTP, HTTPS, FTP, SMTP, webSocket
 2. Each of them has its purposes and functionalities.

Hypertext Transfer Protocol

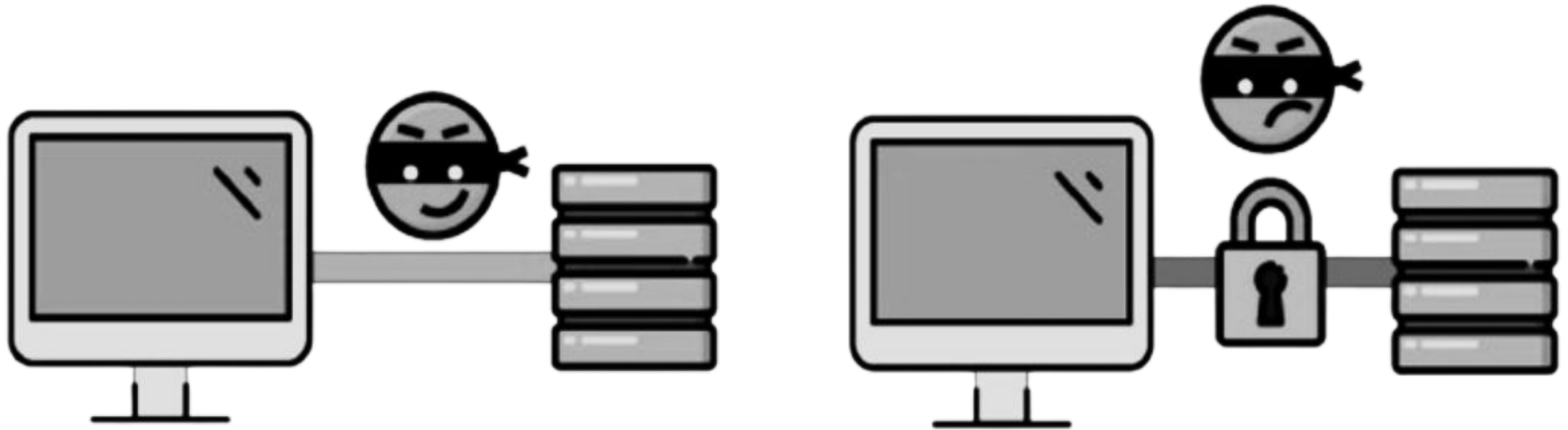
"HTTP"

What is it all about ?

What is HTTP ?

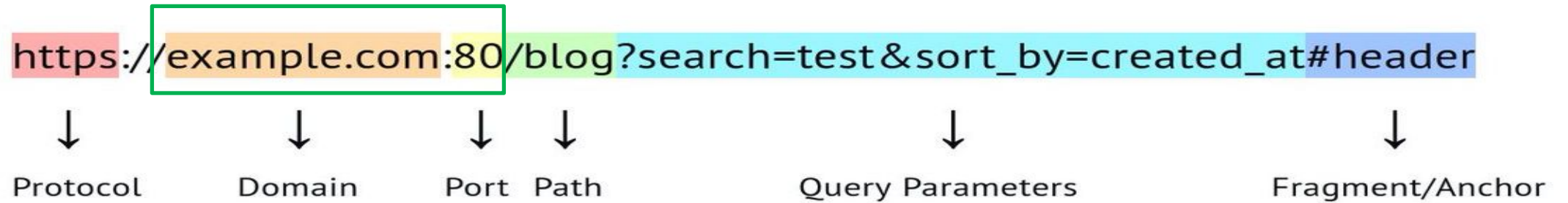
1. HTTP stands for Hypertext Transfer Protocol.
1. It is the standard protocol for transferring web pages (and their content) across the Internet.
1. HTTP provides a structured format for exchanging data over the web.
1. HTTP is a stateless protocol. Server doesn't keep any data between 2 requests.
1. HTTP has the guidelines needed to describe the following:
 1. Client request.
 2. Server response.
2. 6. Let's check the whole request/response cycle....

HTTP vs HTTPS

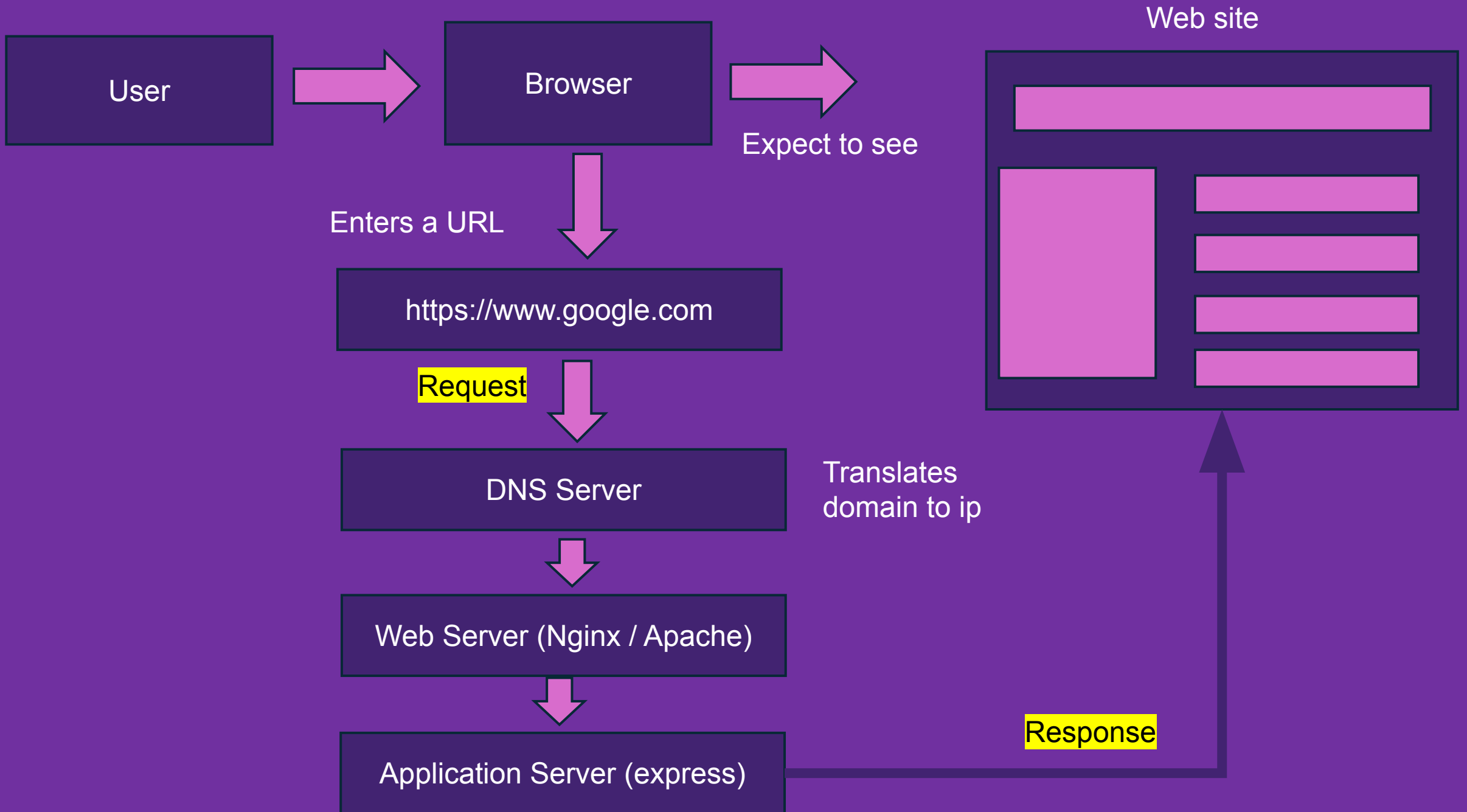


URL:Domain

URL Anatomy



HTTP Request Response Cycle



HTTP General guidelines

1. HTTP request/response usually consists of request/response header and request/response body according to http method.
1. headers hold metadata about the request/response.(http method, accepted format).
1. body usually consists of actual data sent from/to client/server.

HTTP REQUEST

GET /doc/test.html HTTP/1.1

Host: www.test101.com

Accept: image/gif, image/jpeg, */*

Accept-Language: en-us

Accept-Encoding: gzip, deflate

User-Agent: Mozilla/4.0

Content-Length: 35

bookId=12345&author=Tan+Ah+Teck

Request Line

Request Headers

Request
Message
Header

A blank line separates header & body

Request Message Body

HTTP METHODS

- 1. It is a way used by client to describe the type of request (action) being made.
- 1. client may request to create, delete, update or simply read from a resource.
- 1. this corresponds to making POST, DELETE, PUT or GET requests respectively.
- 1. There are other HTTP methods like HEAD, OPTIONS ...etc.

HTTP HEADERS

1. They serve to provide additional information for handling requests and responses.
1. They appear as key value pairs and provide a host of information such as:
 - The cache policy for the response.
 - the acceptable response types enforced by the client.
 - the preferred language of response.
 - Encoding.
 - Credentials for authentication and authorization-such as access tokens.

HTTP RESPONSE

HTTP/1.1 200 OK

Date: Sun, 08 Feb xxxx 01:11:12 GMT

Server: Apache/1.3.29 (Win32)

Last-Modified: Sat, 07 Feb xxxx

ETag: "0-23-4024c3a5"

Accept-Ranges: bytes

Content-Length: 35

Connection: close

Content-Type: text/html

<h1>My Home page</h1>

Status Line

Response Headers

Response
Message
Header

A blank line separates header & body

Response Message Body

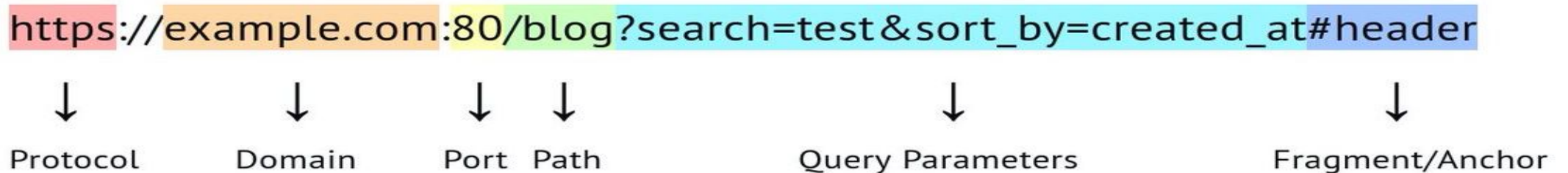
HTTP STATUS CODE

1. It is a useful way to provide information to client about the status of a request.
1. HTTP defines several status codes, each pertaining to a specific scenario.
 - 2xx : imply that the request completed successfully. (200)
 - 3xx : A code in the 3xx series implies redirection. (301)
 - 4xx : A 4xx errors are used when there is an error. (400, 401, 422, 404)
 - 5xx: Lastly, a 5xx response is used when there is an error on the server side. (500)

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URL Anatomy



Thanks